

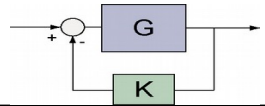
ME 2801

# Introduction to Control Systems

W4: Block Diagrams and Feedback



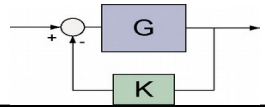
# Plan of the Day



## Reading Quiz

## Block Diagram

- Basic building blocks
- Reduction example, by-hand and with MATLAB
- Feedback example



So Far...

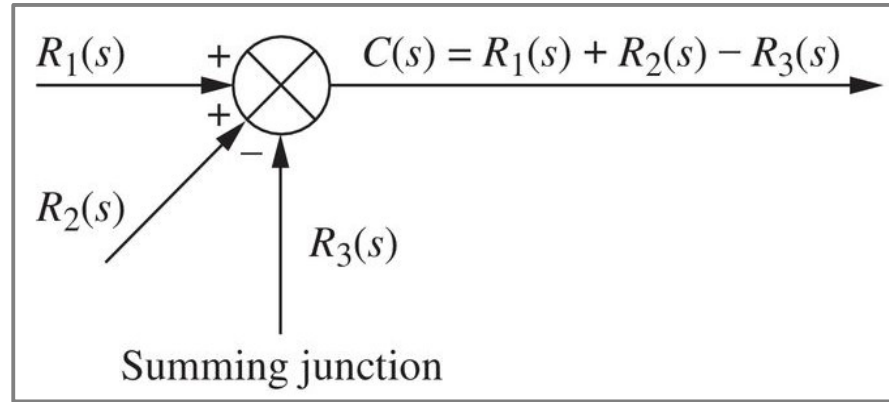
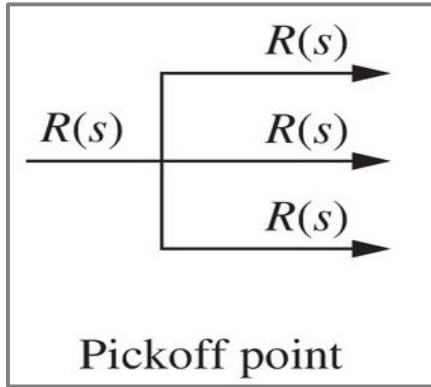
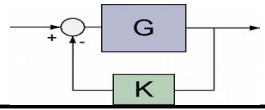
- Laplace transform
- Lumped-parameter models
- First and Second-order response

We can use transfer functions for input-output behavior and s-domain techniques for performance/stability.

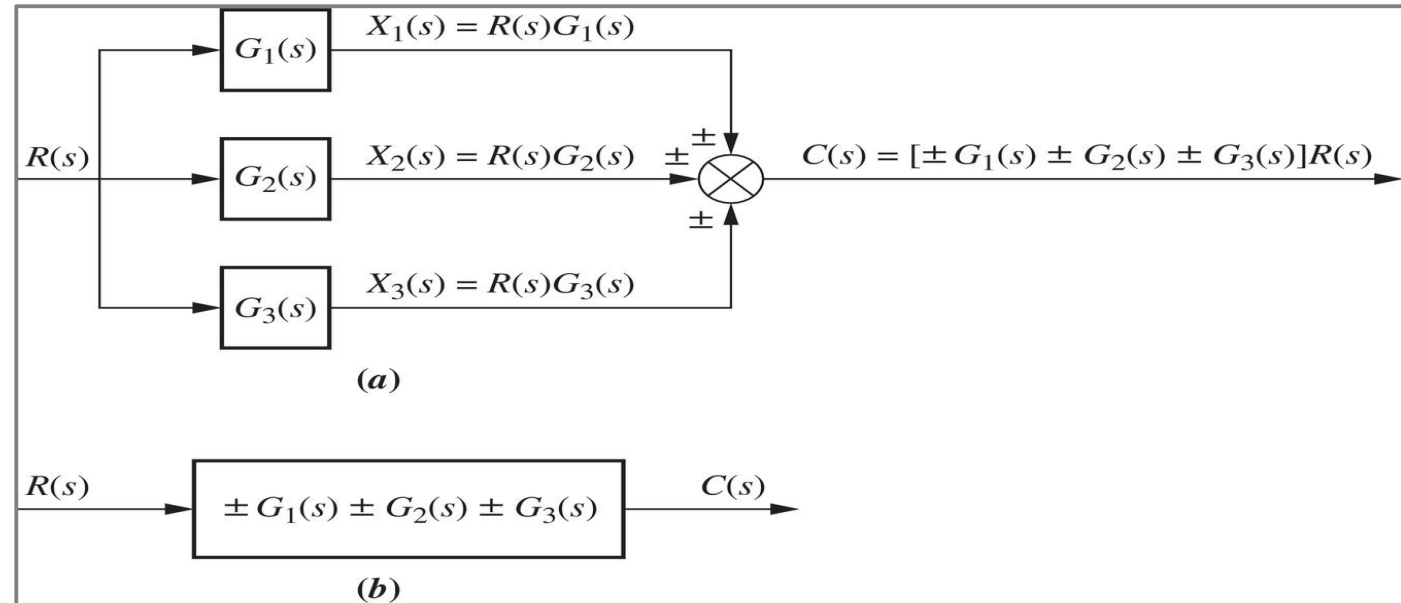
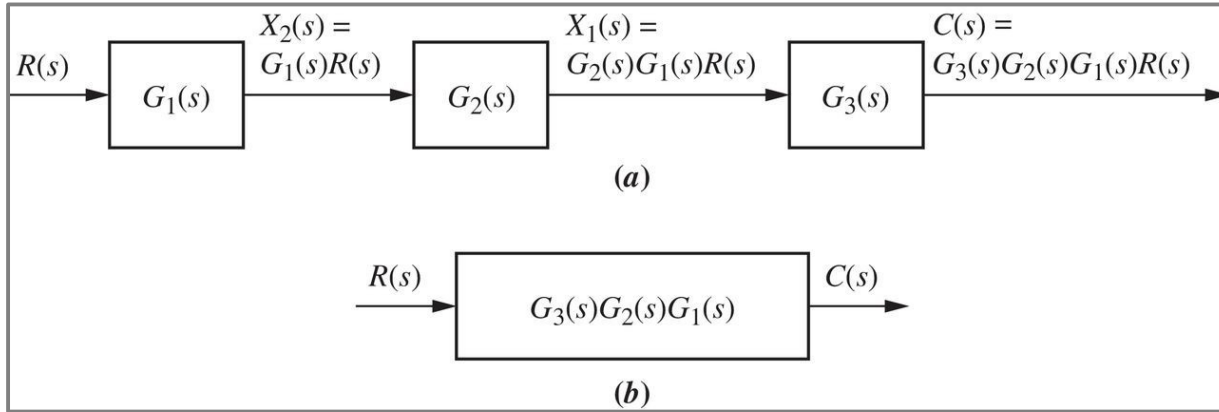
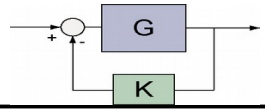
Next...

- Block diagrams allow us to model feedback!
- Predict and design the closed-loop response (performance and stability)

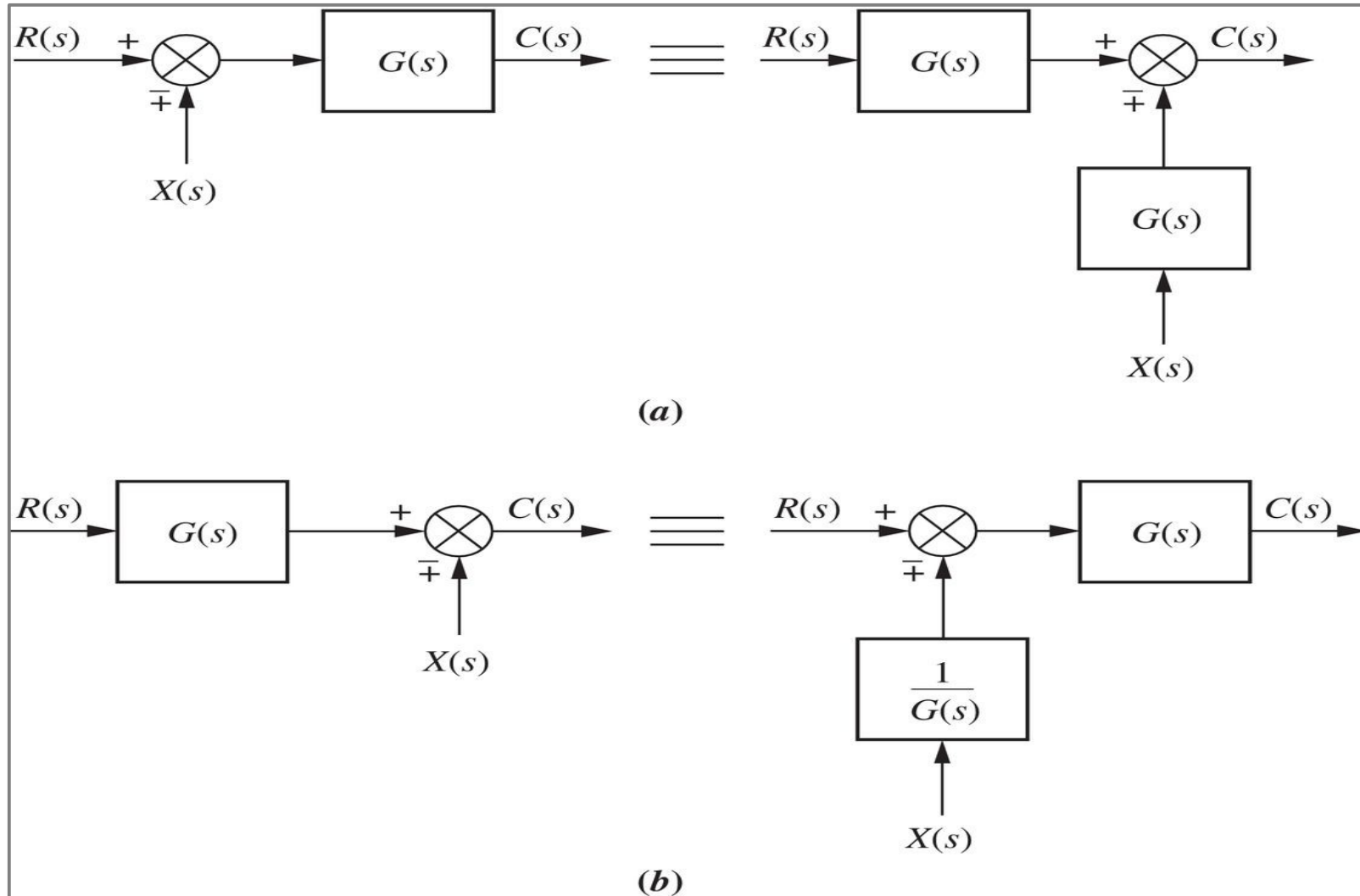
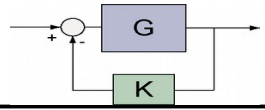
# Summing and Pickoff Junctions



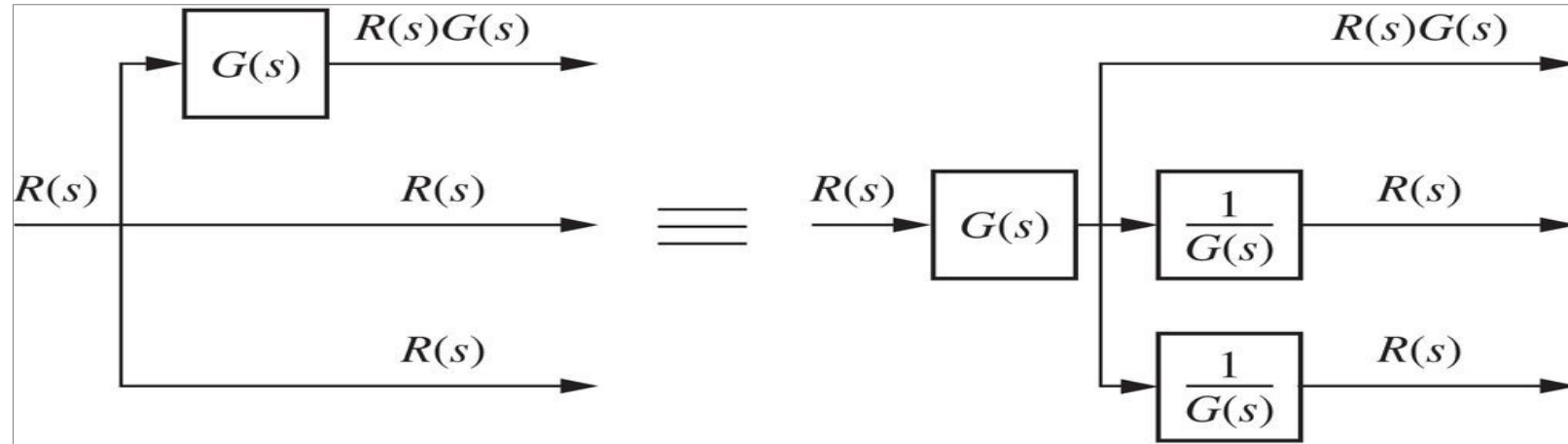
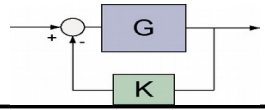
# Series (Cascade) and Parallel



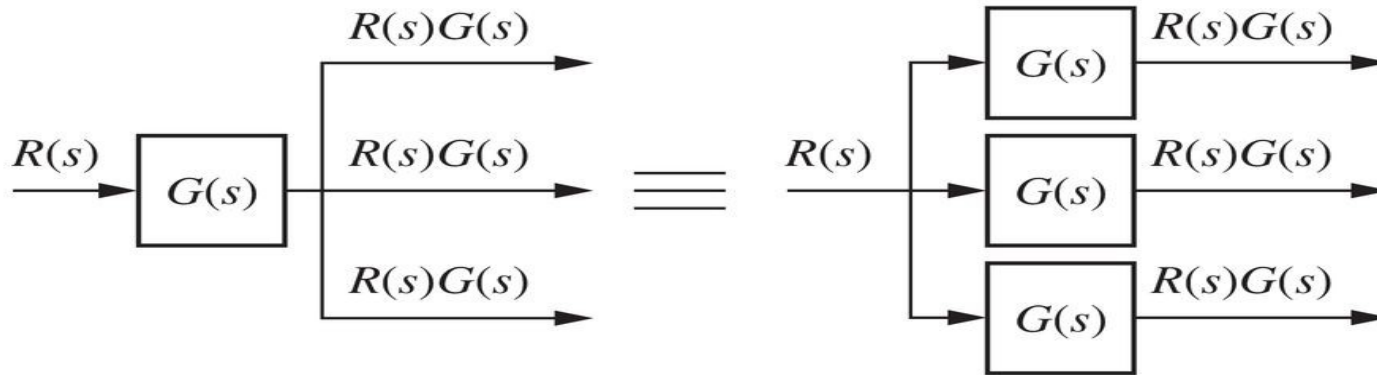
# Moving Across Summing Jctn



# Moving Across Pickoff Jctn

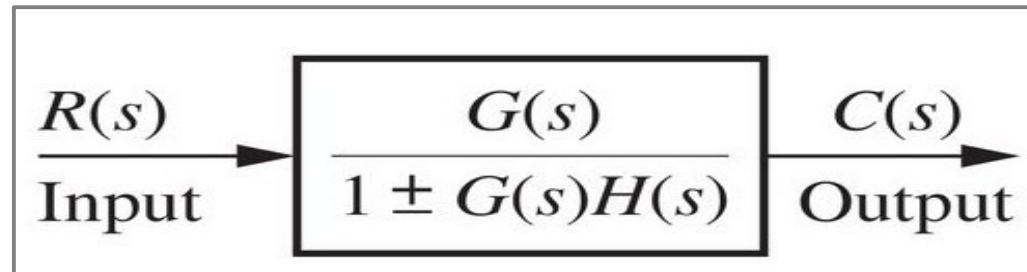
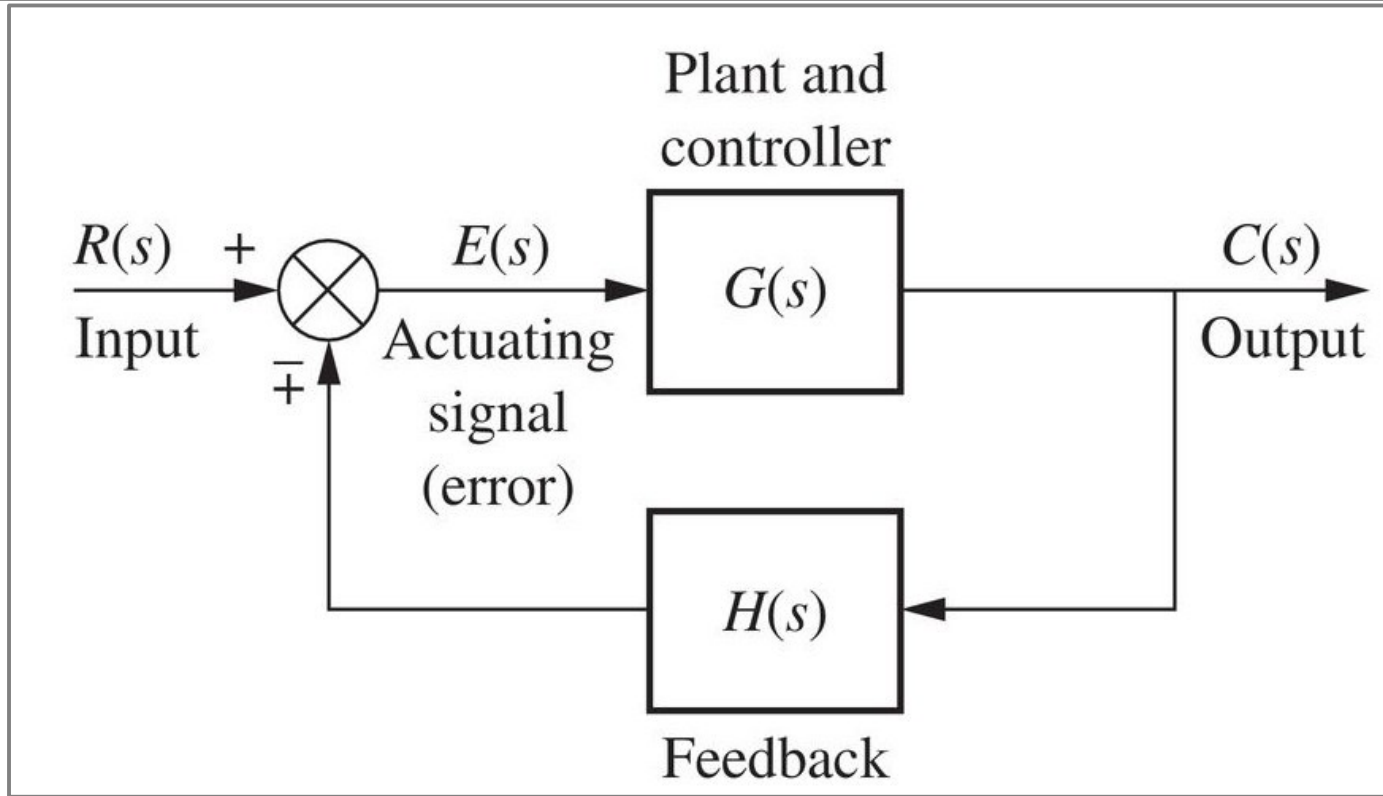
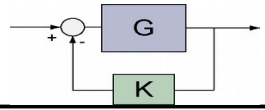


(a)

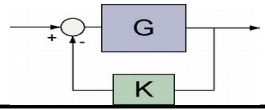


(b)

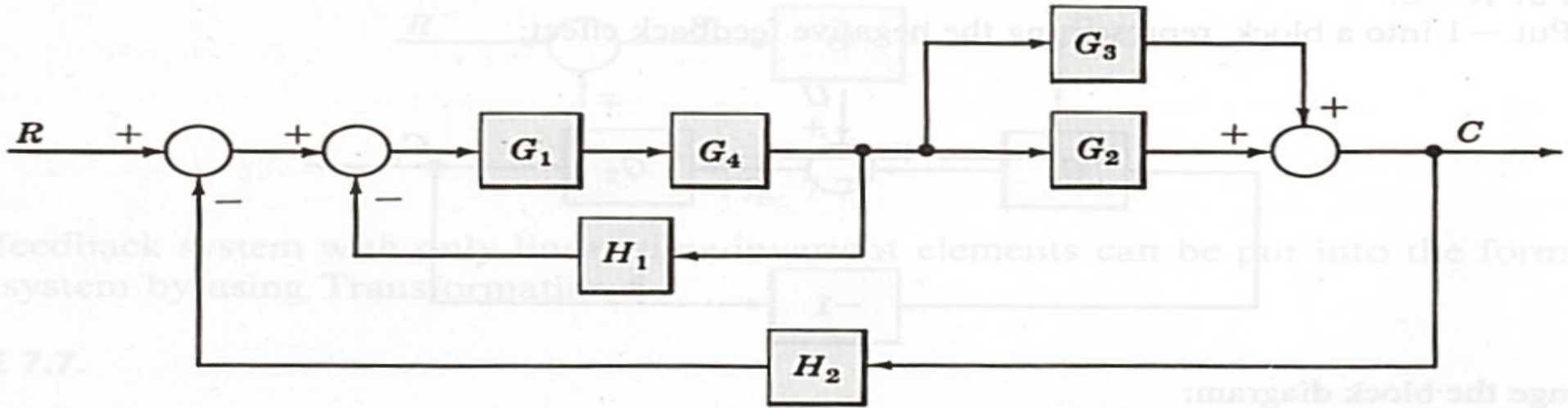
# Feedback Form



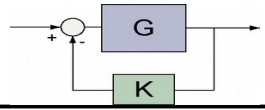
# Example



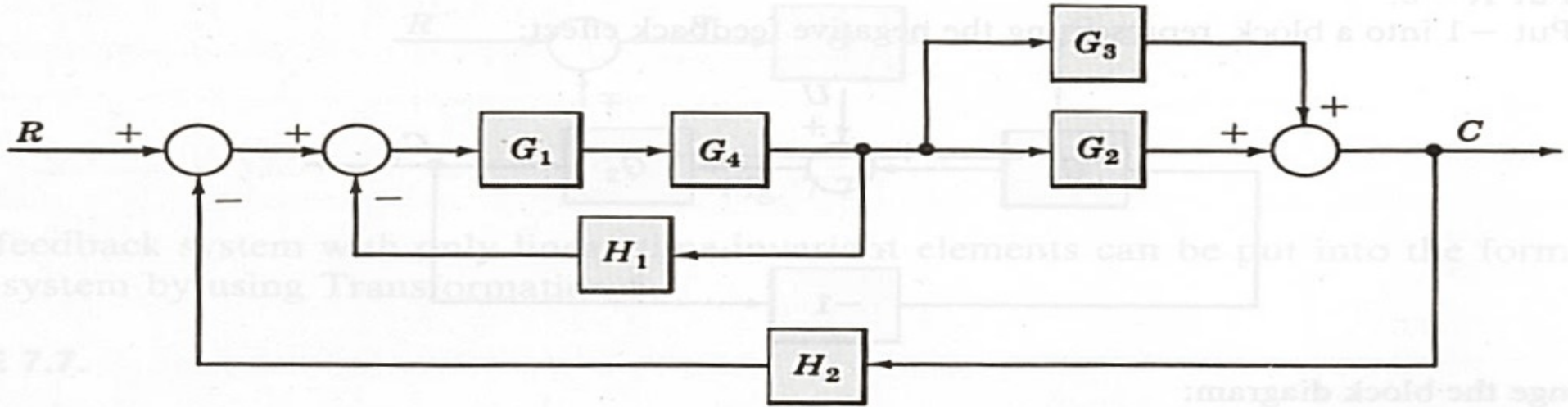
Reduce to single transfer function.



# Example: MATLAB



Reduce to single transfer function.



where

$$G_1 = \frac{1}{(s+1)}$$

$$G_3 = \frac{1}{(s+3)}$$

$$H_1 = \frac{1}{s}$$

$$G_2 = \frac{1}{(s+2)}$$

$$G_4 = \frac{1}{(s+4)}$$

$$H_2 = \frac{1}{s^2}$$

```
G1=tf(1,[1 1]);
```

```
G2=tf(1,[1 2]);
```

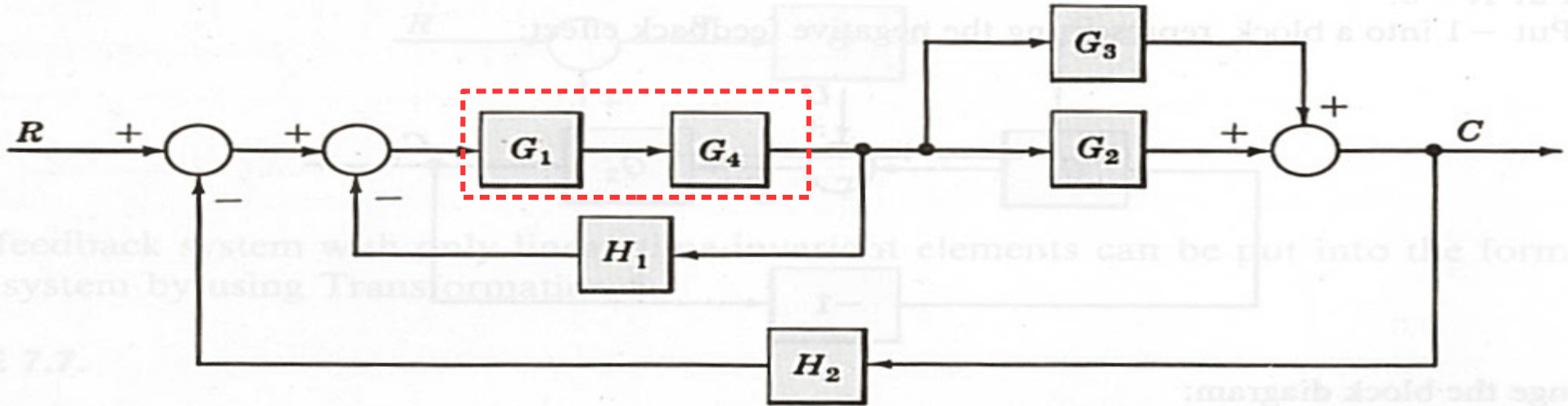
```
G3=tf(1,[1 3]);
```

```
G4=tf(1,[1 4]);
```

```
H1=tf(1,[1 0]);
```

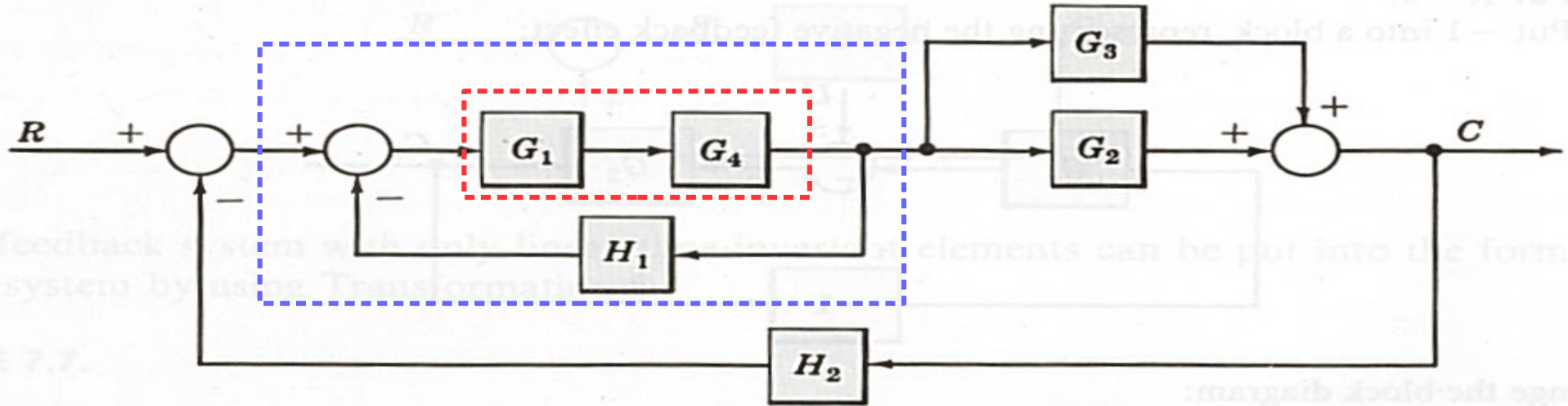
```
H2=tf(1,[1 0 0]);
```

Reduce to single transfer function.



```
G5 = series(G1,G4);  
G5 = G1*G4;
```

Reduce to single transfer function.

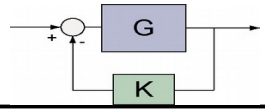


```
G5 = series(G1,G4);
```

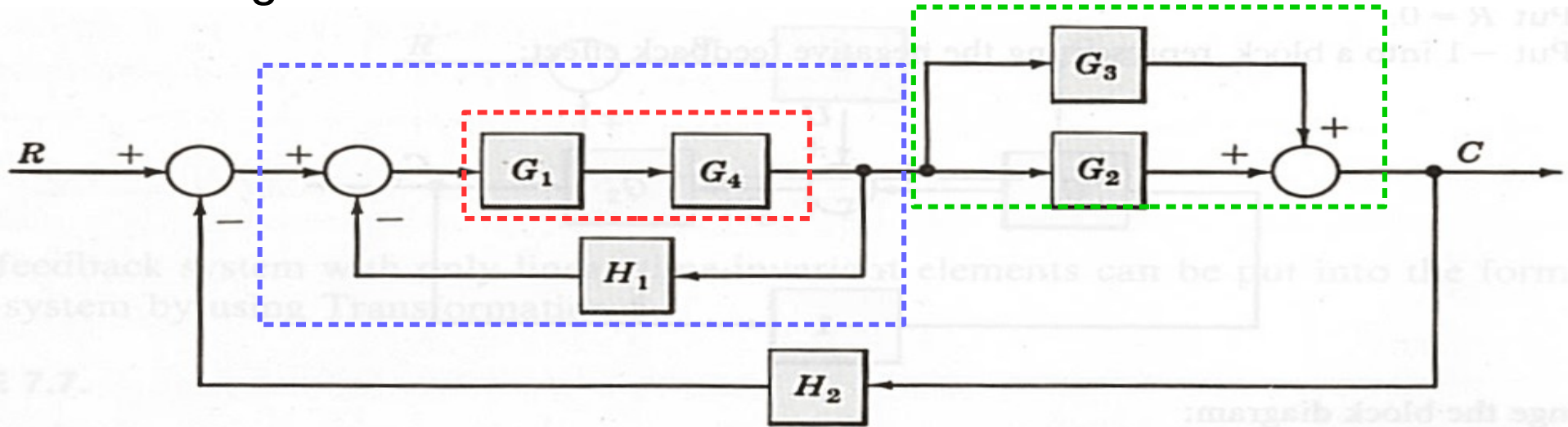
```
G5 = G1*G4;
```

```
G6=feedback(G5,H1);
```

# Example: MATLAB



Reduce to single transfer function.

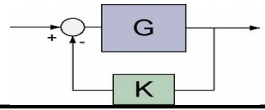


```
G5 = series(G1,G4);
G5 = G1*G4;
```

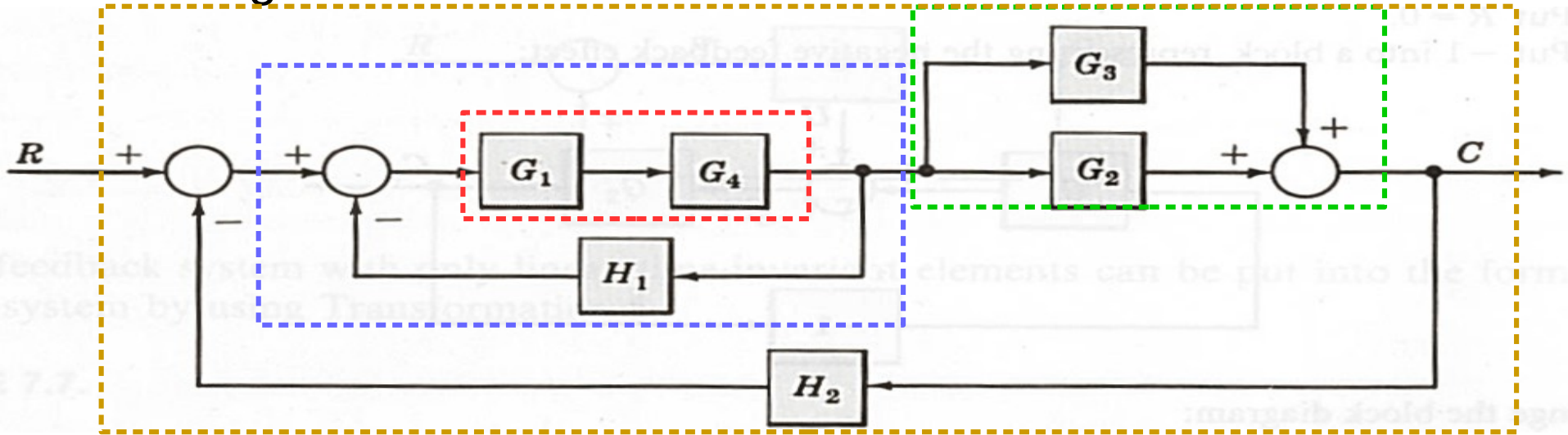
```
G6 = feedback(G5,H1);
```

```
G7 = parallel(G3,G2);
G7 = G3+G2;
```

# Example: MATLAB



Reduce to single transfer function.



```
G5 = series(G1,G4);
```

```
G5 = G1*G4;
```

```
G6 = feedback(G5,H1);
```

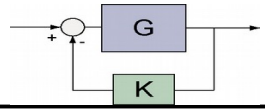
```
G7 = parallel(G3,G2);
```

```
G7 = G3+G2;
```

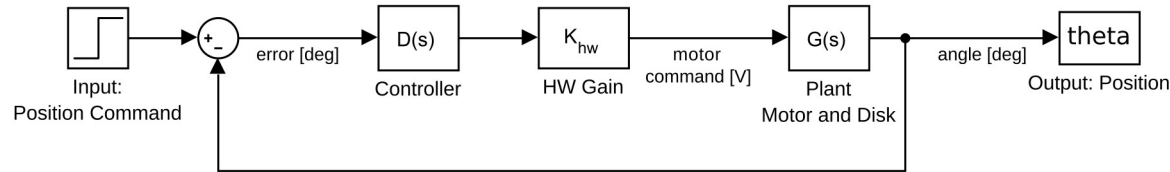
```
G8=feedback(G6*G7,H2);
```

G8 =

$$\frac{2s^4 + 5s^3}{s^7 + 10s^6 + 35s^5 + 51s^4 + 29s^3 + 8s^2 + 5s}$$



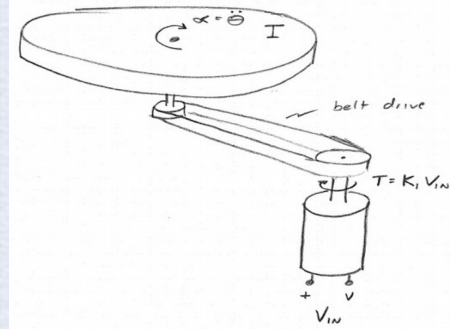
## Given block diagram



$$G(s) = \frac{\theta}{V_{in}} = \frac{K_m}{s(\tau s + 1)}$$

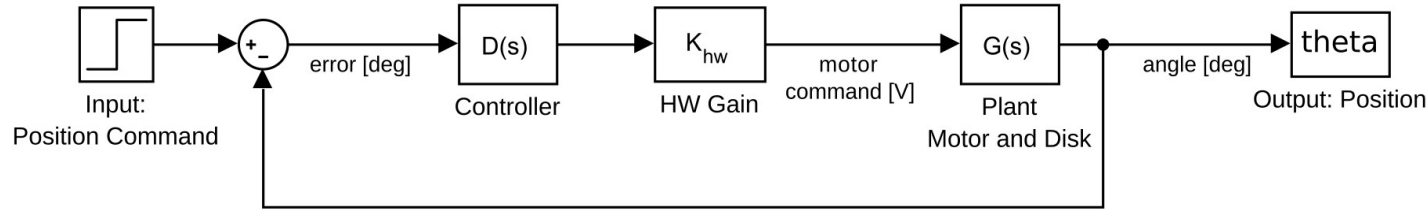
Find closed-loop diagram for proportional control,  $D(s) = K_p$

- Predict the performance (settling time, peak time, %OS, rise time, etc.) as a function of the proportional gain.



Given block diagram

$$G(s) = \frac{\theta}{V} = \frac{K_m}{s(\tau s + 1)}$$



Find closed-loop diagram for proportional control

$$D(s) = K_p$$

- Predict the performance (settling time, peak time, %OS, rise time, etc.) as a function of the proportional gain.

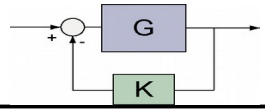
$$G_e(s) = \frac{\theta}{V} = \frac{(K_p K_{hw} K_m) / \tau}{s^2 + (1/\tau)s + (K_p K_{hw} K_m) / \tau}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$



## Closed-Loop Transfer Function

$$G_e(s) = \frac{\theta}{V} = \frac{(K_p K_{hw} K_m) / \tau}{s^2 + (1/\tau)s + (K_p K_{hw} K_m) / \tau}$$

```
% ECP Parameters
```

```
Khw = 0.434;
```

```
Km = 25000;
```

```
tau = 4;
```

```
% Proportional Gain
```

```
Kp = 0.0001;
```

```
% By Hand
```

```
Ge = tf(Kp*Km*Khw/tau,...  
        [1 1/tau Kp*Km*Khw/tau])
```

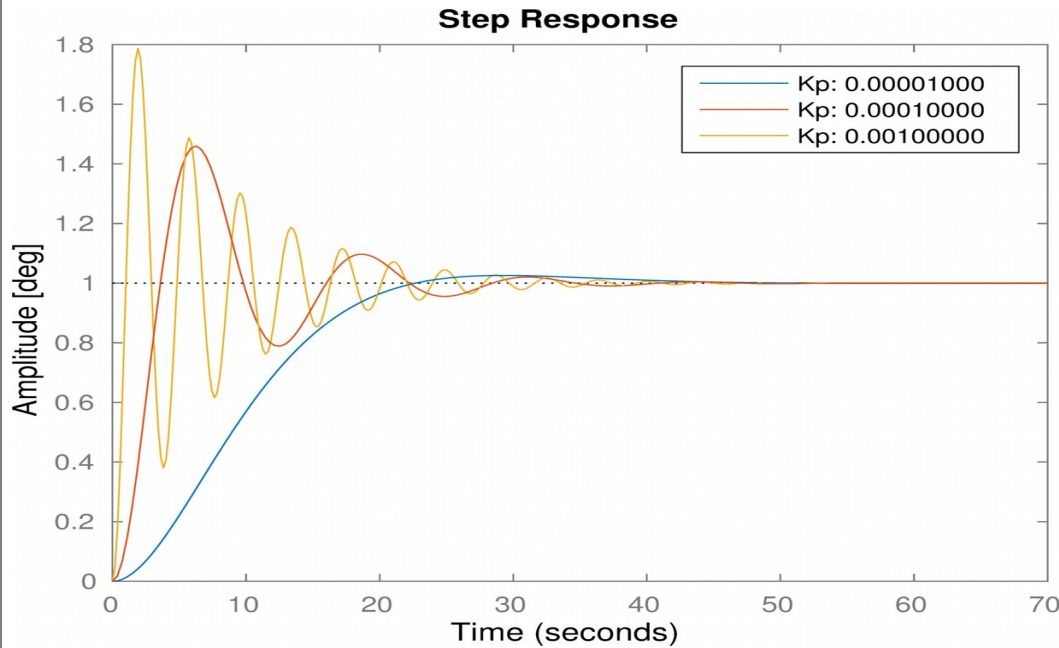
```
% Using MATLAB block diagram algebra
```

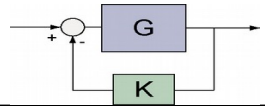
```
G = tf(Km,[tau 1 0]); % ECP plant
```

```
Ge2 = feedback(Kp*Khw*G,1)
```

```
figure(1); clf();
```

```
step(Ge,Ge2)
```





## Block Diagrams

- Using visual algebra to manipulate block diagram system descriptions
- Basic forms
  - Summation, pickoff, cascade (series) and parallel
  - Negative feedback

## Feedback Design

- Combine
  - Block diagram algebra
  - Transient performance metrics
- Predict the system performance (dynamics) based on simple models of feedback control.